

Multi-Omics Factor Analysis

Integrative Analysis with CCA and MOFA2

Agenda

Multi-omics factor analysis - integration challenges

Canonical Correlation Analysis (CCA)

Multi-Omics Factor Analysis (MOFA2)

The Integration Problem

Each omics layer measures different molecular entities:

- Different scales (counts, continuous, binary)

- Different dimensions (thousands of features)

- Different noise characteristics

Traditional approaches (QTL, WGCNA) discover simple connections

In reality - multiple features participate/work together

Solution: Multi-omics factor analysis frameworks - find shared patterns (latent factors) that explain variation across all omics layers

What is a Latent Factor?

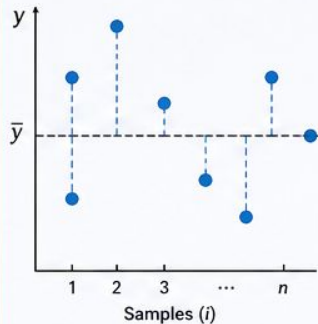
- A latent factor is a hidden (unobserved) variable inferred from data
- It is a linear combination of observed variables (e.g., genes, SNPs, features)
- Capture (explain) **variation** across multiple variables
- Used for dimensionality reduction (many variables → few factors)
- Each factor has:
 - Factor scores (per sample)
 - Loadings / weights (per variable)
- Helps reveal underlying biological or statistical structure
- Examples:
 - BMI
 - Inflammation level: Gene expression of IL6, TNF, CRP + White blood cell count:
$$\text{Factor1} = 0.5 \cdot \text{IL6} + 0.4 \cdot \text{TNF} + 0.6 \cdot \text{CRP} + 0.3 \cdot \text{WBC}$$

Explained Variance, Unexplained Variance and R²

How much of the variation in the data is captured by the model?

1. Variation (Variance)

Total variation is the spread of the observed values around their mean.



Mean:

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

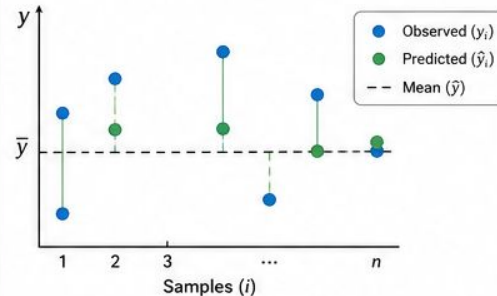
Total Sum of Squares (TSS):

$$TSS = \sum_{i=1}^n (y_i - \bar{y})^2$$

This is the total (baseline) variability in the data.

2. Model Predictions

The model produces predicted values $\hat{y}_i$.



Blue dots = true values (y_i)
Green dots = model predictions ($\hat{y}_i$)
Dashed line = mean of y

3. Decomposition of Variation

Total variation (TSS) can be split into two parts.

$$TSS = ESS + RSS$$

Explained Variation (ESS)

Variation explained by the model

$$ESS = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

How much the predictions deviate from the mean.

Unexplained Variation (RSS)

Variation NOT explained by the model (error)

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

How much the true values deviate from the predictions.

Interpretation

TSS = total spread, ESS = signal captured, RSS = noise / missing structure

4. Concrete Example

Observed values (y)

Predicted values ($\hat{y}$)

i	1	2	3
y_i	160	170	180

$\hat{y}_i$	162	168	178
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Step 1: Mean of y

$$\bar{y} = (160 + 170 + 180) = 170$$

Step 2: Total variance (TSS)

$$TSS = (160-170)^2 + (170-170)^2 + (180-170)^2 = 100 + 0 + 100 = 200$$

Step 4: Explained variance (ESS)

$$ESS = TSS - RSS = 200 - 12 = 188$$

Step 5: R²

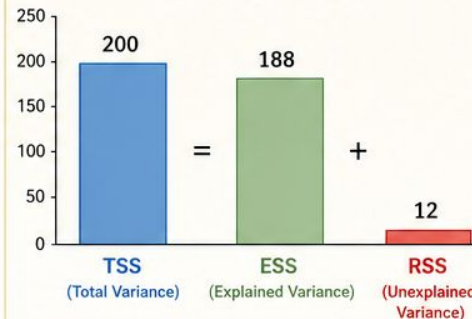
$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{12}{200} = 0.94 \quad (94\% \text{ explained})$$

Step 3: Unexplained variance (RSS)

$$RSS = (160-162)^2 + (170-168)^2 + (180-178)^2 = 4 + 4 + 4 = 12$$

5. Visual Decomposition

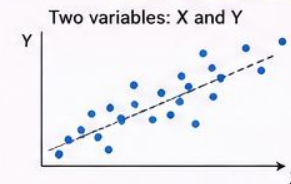
$$TSS = ESS + RSS$$



$$R^2 = ESS/TSS = 188/200 = 0.94$$

→ 94% of the variation is explained by the model.
6% is unexplained.

6. Shared Variation (Two Variables)

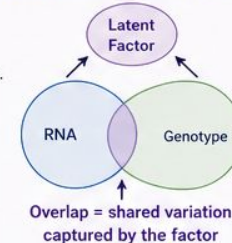


If correlation $r = 0.70$
 $R^2 = r^2 = 0.49$
→ 49% of the variation in Y is shared (explained) by X .
51% is independent (not shared / noise / other factors).

7. Connection to Latent Factors (e.g., CCA, MOFA)

Single dataset
A latent factor explains variance within one dataset.

Example:
Factor 1 explains 30% of variance in RNA



Multiple datasets
A latent factor explains shared variance across datasets.

Example:
Same factor explains 20% in Genotype and 30% in RNA

Key Takeaways



Variance (TSS)

Total spread of the data around the mean.

Explained Variance (ESS)

Variation captured by the model (signal).

Unexplained Variance (RSS)

Variation not captured (noise / missing structure).

$$R^2 = ESS / TSS$$

Fraction of total variance explained by the model.

Shared Variation

The overlap in variation between variables or datasets.

Canonical Correlation Analysis (CCA)

Classical method for relating two datasets (developed 1936)

How CCA works:

- Finds linear combinations of features in each dataset

- Maximizes correlation between the two combinations

- Result: Pairs of 'canonical variates' that are maximally correlated

Key properties:

- Forces both datasets to contribute equally

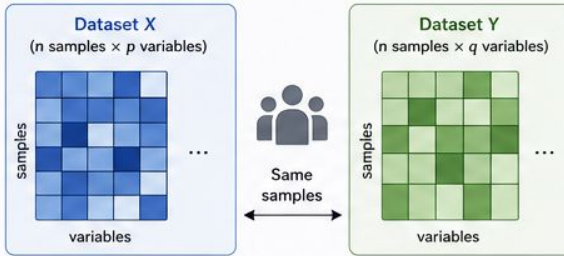
- Works well for two-view integration

- No regularization (can overfit with small number of samples)

Canonical Correlation Analysis (CCA) – Under the Hood

Finds linear combinations of two datasets that are maximally correlated.

1. THE SETUP



Goal: find linear combinations (canonical variates) of X and Y that are maximally correlated.

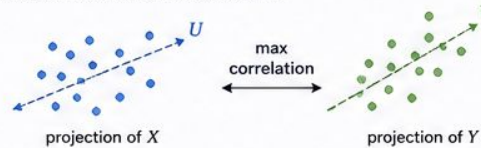
2. LINEAR COMBINATIONS (CANONICAL VARIATES)

Find weight vectors a and b :

$$U = Xa \quad (n \times 1) \quad \text{first canonical variate for X}$$
$$V = Yb \quad (n \times 1) \quad \text{first canonical variate for Y}$$

$\text{cor}(U, V)$ is maximized

Intuition: project X and Y into 1D so that the projections (U and V) are as correlated as possible.



3. MATHEMATICAL FORMULATION

We solve:

$$\max_{a,b} \text{corr}(Xa, Yb) = \frac{a^T \Sigma_{XY} b}{\sqrt{a^T \Sigma_{XX} a} \sqrt{b^T \Sigma_{YY} b}}$$

Subject to: $a^T \Sigma_{XX} a = 1$ and $b^T \Sigma_{YY} b = 1$

Where:

$$\Sigma_{XX} = \frac{1}{n-1} X^T X \quad (\text{covariance of X})$$
$$\Sigma_{YY} = \frac{1}{n-1} Y^T Y \quad (\text{covariance of Y})$$
$$\Sigma_{XY} = \frac{1}{n-1} X^T Y \quad (\text{cross-covariance})$$

This is a generalized eigenvalue problem.

The solutions give a and b (weight vectors) that maximize correlation.

4. THE ALGORITHM (FIRST PAIR)

- Standardize (center) X and Y.
- Compute covariance matrices Σ_{XX} , Σ_{YY} , Σ_{XY} .
- Solve the generalized eigenvalue problem:

$$\Sigma_{XX}^{-1} \Sigma_{XY} \Sigma_{YY}^{-1} \Sigma_{YX} a = \lambda a$$

$$\Sigma_{YY}^{-1} \Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY} b = \lambda b$$

- Take the eigenvector for the largest eigenvalue λ_1 .
- Normalize a and b :

$$a \leftarrow \frac{a}{\sqrt{a^T \Sigma_{XX} a}}$$

$$b \leftarrow \frac{b}{\sqrt{b^T \Sigma_{YY} b}}$$

- Compute canonical variates: $U = Xa$, $V = Yb$.
- The canonical correlation is $\rho_1 = \text{corr}(U, V) = \sqrt{\lambda_1}$.

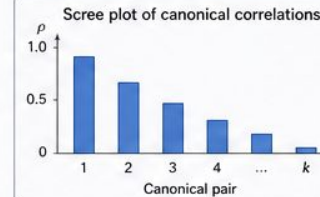
Further pairs: Remove the effect of previous pairs (deflation) and repeat to get (a_2, b_2) , (a_3, b_3) , ... with $\rho_1 \geq \rho_2 \geq \dots \geq \rho_k$

5. MULTIPLE CANONICAL PAIRS

CCA finds k pairs: (U_1, V_1) , (U_2, V_2) , ..., (U_k, V_k)

Properties:

- $\text{corr}(U_i, V_i) = \rho_i$
- U_i is uncorrelated with U_j for $i \neq j$ (under Σ_{XX})
- V_i is uncorrelated with V_j for $i \neq j$ (under Σ_{YY})
- Pairs are ordered by decreasing correlation: $\rho_1 \geq \rho_2 \geq \dots \geq \rho_k$



Interpretation



High ρ_i
Strong shared variation captured by the i -th pair



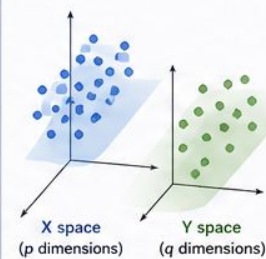
Weights (a_i, b_i) show which original variables contribute most



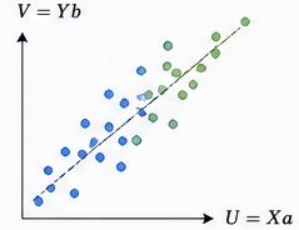
Useful for integration, dimensionality reduction, and feature discovery

6. GEOMETRIC VIEW

Original high-dimensional spaces



After CCA projection



CCA finds directions (a and b) so that the projections are maximally correlated.

7. SMALL NUMERICAL EXAMPLE

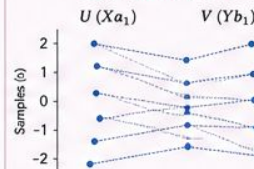
Suppose after computation we get:

Pair i	ρ_i	Interpretation
1	0.87	Strong shared signal
2	0.52	Moderate shared signal
3	0.23	Weak shared signal

Example weights (first pair)

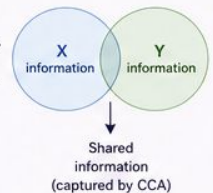
Top variables in X (a_1)		Top variables in Y (b_1)	
Gene_15	0.42	SNP_22	-0.45
Gene_8	-0.31	SNP_7	0.33
Gene_3	0.28	SNP_103	0.27
...		...	

Canonical variates



Key Takeaways

- CCA finds linear combinations of two datasets that are maximally correlated.
- It reveals shared structure across two different views of the same samples.
- Output: canonical correlations (ρ_i) and weights (a_i, b_i).
- Powerful for multi-omics integration, genotype-expression analysis, and uncovering biological signals.



CCA: Mathematical Framework

Given two data matrices:

X : n samples $\times$ p features

Y : n samples $\times$ q features

CCA finds weight vectors (w_x, w_y):

Maximize: $\text{corr}(Xw_x, Yw_y)$

Optimization Problem:

$\max \text{corr}(Xw_x, Yw_y)$

subject to:

$\text{var}(Xw_x) = 1$

$\text{var}(Yw_y) = 1$

Result: Canonical variates

$U = Xw_x$ (from first dataset)

$V = Yw_y$ (from second dataset)

U and V are maximally correlated

Calculates factor vectors

$$\text{corr}(Xw_x, Yw_y) = \frac{\text{cov}(Xw_x, Yw_y)}{\sqrt{\text{var}(Xw_x) \text{var}(Yw_y)}}$$

Variance 1 is just a convenient normalization constant - any fixed value would work, but choosing 1 makes the math clean and interprets the vectors as unit-length.

CCA: How Weights Are Calculated

Optimization algorithm: Eigenvalue decomposition

Step-by-step process:

1. Standardize both datasets

Center (mean=0) and scale (std=1)

2. Compute covariance matrices:

$C_{XX} = \text{cov}(X, X)$ - within view 1

$C_{YY} = \text{cov}(Y, Y)$ - within view 2

$C_{XY} = \text{cov}(X, Y)$ - between views

3. Solve generalized eigenvalue problem:

Find weights that maximize correlation

Uses matrix decomposition

Mathematical Solution:

Step 1: Compute cross-covariance

$$C_{XY} = X^T \times Y / n$$

Step 2: Solve eigenvalue equation

$$C_{XX}^{-1} \times C_{XY} \times C_{YY}^{-1} \times C_{YX} \times w = \lambda^2 w$$

Step 3: Extract weights

Eigenvector with largest eigenvalue
→ gives first pair of weights

Step 4: Repeat for more factors

Find orthogonal directions
→ subsequent canonical variates

Fast computation: $O(p^3)$ complexity
where p = number of features

CCA Example: Step-by-Step

Toy Dataset Example:

300 samples (patients)

View 1: 10 gene expression values

View 2: 50 SNP genotypes (0/1/2)

CCA Process:

1. Standardize both datasets

Mean = 0, Std = 1 for each feature

2. Find linear combinations:

$$\text{Factor_score}_1 = w_1 \cdot \text{gene}_1 + w_2 \cdot \text{gene}_2 + \dots + w_{10} \cdot \text{gene}_{10}$$

$$\text{Factor_score}_2 = u_1 \cdot \text{SNP}_1 + u_2 \cdot \text{SNP}_2 + \dots + u_{50} \cdot \text{SNP}_{50}$$

Linear Combination Example:

Expression view weights:

```
gene_1: 0.23
gene_2: -0.18
gene_3: 0.31
...
gene_10: 0.12
```

Genotype view weights:

```
SNP_1: 0.08
SNP_2: -0.15
SNP_3: 0.22
...
SNP_50: 0.09
```

Each sample gets a factor score:

```
Sample_1: Factor = 1.43
Sample_2: Factor = -0.87
...
```

CCA Limitations

Limited to two datasets (views):

- Cannot integrate genomics + transcriptomics + proteomics simultaneously

- Requires pairwise comparisons for >2 datasets

No regularization:

- Prone to overfitting with small sample sizes

- All features weighted equally (no sparsity)

Forces equal contribution:

- Both views must contribute even if signal is primarily in one

- Can capture noise when forcing integration

Multi-Omics Factor Analysis (MOFA2)

Modern Bayesian framework for multi-omics integration

Developed by Biofam group (biofam.github.io/MOFA2)

Key advantages over CCA:

- Handles any number of omics layers (views) simultaneously

- Automatic regularization prevents overfitting

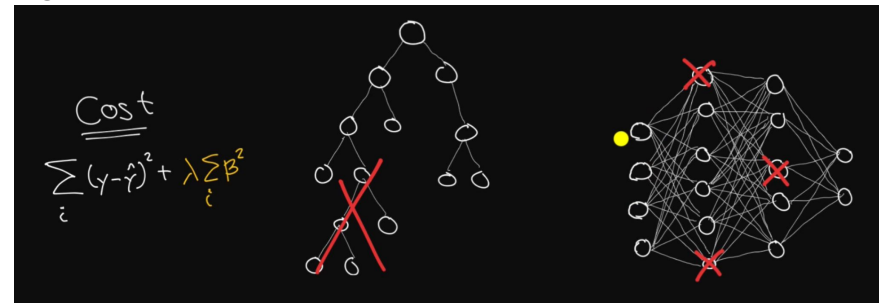
- Identifies sparse factor loadings (feature selection)

- Flexible likelihoods for different data types

- Quantifies uncertainty (Bayesian approach)

Think of MOFA2 as:

- Multi-view PCA with automatic sparsity and regularization



MOFA2: How It Works

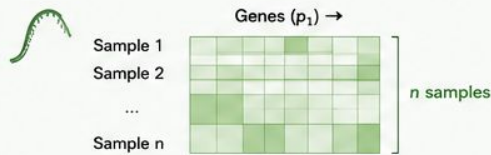
Integrating multiple data types measured on the same set of samples

1. INPUT: Multiple data matrices (views)

All views measured on the same n samples

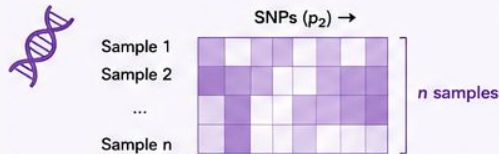
View 1: Expression data

(n samples $\times$ p_1 genes)



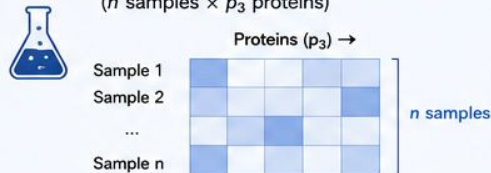
View 2: Genotype data

(n samples $\times$ p_2 SNPs)



View 3: Clinical data

(n samples $\times$ p_3 proteins)

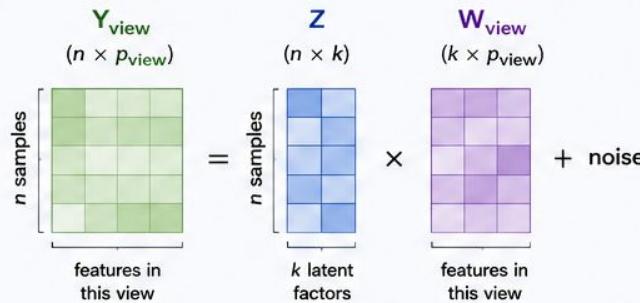


⋮
... more views

2. MODEL ASSUMPTION

Each view is generated from
the same set of hidden (latent) factors

$$Y_{\text{view}} = Z \times W_{\text{view}} + \text{noise}$$



Z: Shared factors

(n samples $\times$ k factors)

Captures hidden sources of
variation that affect
ALL views

W_{view} : View-specific weights

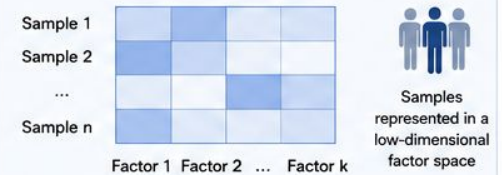
(k factors $\times$ features)

Tells how strongly each
factor influences each
feature in this view

3. WHAT MOFA2 LEARNS from the data

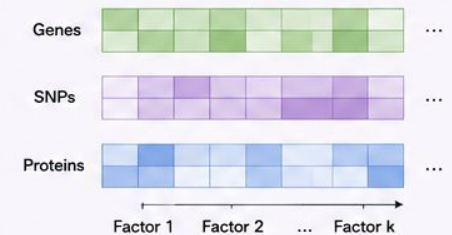
A. Latent factors (Z)

Shared sources of variation
across ALL views



B. Factor loadings (W_{view})

Which features drive each factor
in each view



In simple terms: MOFA2 finds a small set of **hidden factors (Z)** that explain patterns in all your different data types. It also tells you which genes, SNPs, proteins, etc. are **important for each factor (W)**.

MOFA2: Algorithm Details

Step 1 - Initialization:

- Start with k factors (user-defined or automatic)

- Initialize factor values and weights randomly

Step 2 - Expectation-Maximization (EM) iterations:

- E-step: Update factor values Z given current weights

- M-step: Update weights W given current factors

Step 3 - Regularization (Bayesian priors):

- Spike-and-slab prior: Drives many weights to exactly zero (sparsity)

- ARD prior: Automatically removes uninformative factors

Step 4 - Convergence:

- Iterate until change in variance explained $<$ threshold (Evidence Lower Bound)

- Drop factors explaining $< 1\%$ variance

MOFA2 Example: Linear Combination with Sparsity

Same dataset: 300 samples

View 1: 10 genes

View 2: 50 SNPs

MOFA2 learns sparse weights:

Most weights = 0 (automatic feature selection)

Factor calculation (per sample):

Expression contribution:

Factor = $w_1 \times \text{gene}_1 + w_2 \times \text{gene}_2 + \dots + w_{10} \times \text{gene}_{10}$

Genotype contribution:

Factor = $u_1 \times \text{SNP}_1 + u_2 \times \text{SNP}_2 + \dots + u_{50} \times \text{SNP}_{50}$

Sparse Weight Example (Factor 1):

Expression view:

```
gene_1: 0.00 ← Sparse (zero)
gene_2: 0.00 ← Sparse (zero)
gene_3: 0.87 ← Active!
gene_4: 0.00 ← Sparse (zero)
gene_5: 0.43 ← Active!
gene_6: 0.00 ← Sparse (zero)
...
gene_10: 0.21 ← Active!
```

Only 3/10 genes have non-zero weights!

Genotype view:

```
SNP_1 to SNP_50: ALL 0.00
→ This factor ignores genotypes
```

Sample score: $0.87 \times \text{gene}_3 + 0.43 \times \text{gene}_5 + 0.21 \times \text{gene}_{10}$

This factor explains variation only in gene expression!

MOFA2: Key Features for Biologists

Automatic feature selection:

- Identifies which genes/SNPs/proteins drive each factor

- Most weights are exactly zero (sparse loadings)

Variance explained:

- Quantifies how much variation each factor captures per view

- Helps prioritize factors for biological interpretation

Factor interpretation:

- Factors can be correlated with phenotypes (disease status, age, etc.)

- View-specific loadings reveal molecular drivers

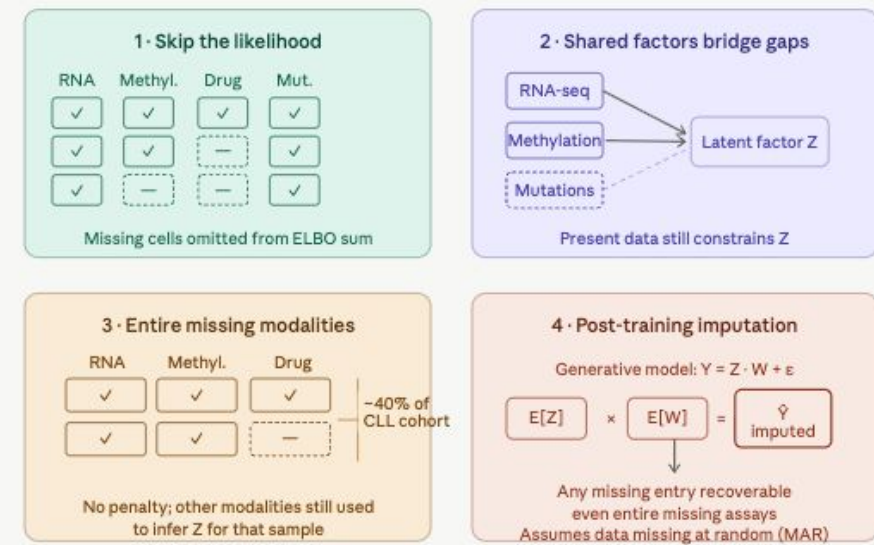
Handles missing data:

- Not all samples need measurements in all views

- Useful for clinical datasets with incomplete measurements

How MOFA handles missing data?

1. Skip the likelihood - the core mechanism. Missing cells are simply excluded from the ELBO objective sum
2. Shared factors bridge gaps - because latent factors Z are shared across modalities, a sample with partial data still contributes to factor inference through whatever modalities are observed
3. Entire missing modalities - handles the common cohort scenario where a whole assay is absent for a subset of patients
4. Post-training imputation - once training is done, any missing entry can be recovered as $\hat{E}[Y] = E[Z] \cdot E[W]$. This is a downstream use of the trained model, not part of the inference itself. It rests on the missing-at-random (MAR) assumption.



MOFA vs MOFA2

Aspect	MOFA (2018)	MOFA+ / MOFA2 (2020)
Core goal	Unsupervised integration of multi-omics (primarily bulk)	Generalized framework for multi-modal + single-cell + multi-group data
Inference method	Variational inference (early implementation, less scalable)	Improved stochastic variational inference → scalable to large datasets
Scalability	Moderate (hundreds–thousands of samples)	High (thousands–millions of single cells)
Data type focus	Bulk multi-omics	Single-cell + bulk + multi-modal
Handling missing data	Supported	More robust handling of missing modalities (very common in single-cell)
Sparsity constraints	Basic sparsity	Flexible sparsity (ARD, spike-and-slab–like behavior) → better interpretability
Likelihood models	Mainly Gaussian	Multiple likelihoods (Gaussian, Poisson, Bernoulli, etc.) for different data types
Heterogeneous modalities	Supported but less flexible	Explicitly designed for heterogeneous modalities (RNA, ATAC, methylation, etc.)
Computational efficiency	Slower on large/high-dimensional data	Significantly faster and more memory-efficient
Implementation	Original MOFA (R/Python)	MOFA2 package (rewritten, optimized backend)

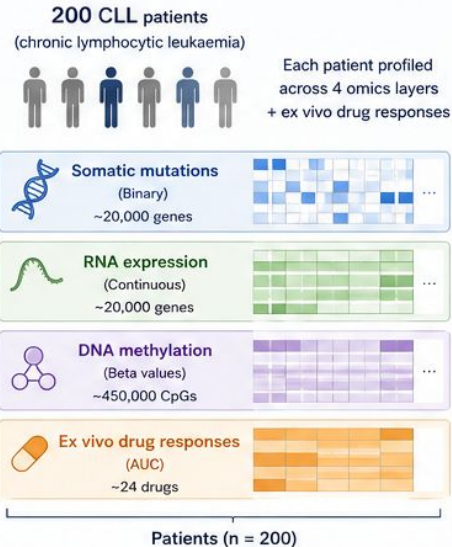
MOFA - original publication

Multi-Omics Factor Analysis (MOFA): Unsupervised Integration of Multi-Omics Data

Argelaguet et al., Molecular Systems Biology (2018) | ~1,000+ citations

MOFA is a flexible Bayesian factor analysis framework that uncovers latent sources of variation shared across multiple omics layers.

STUDY DESIGN & DATA

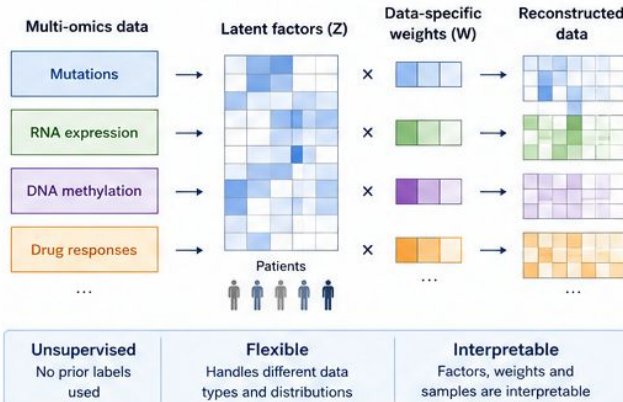


KEY CONTRIBUTIONS

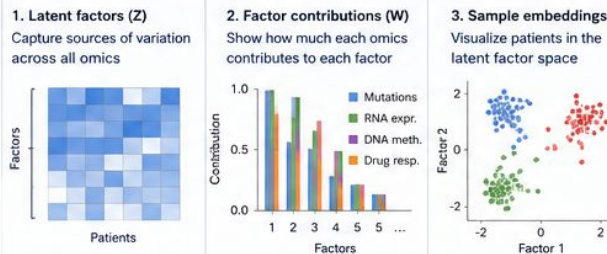
- ✓ First unsupervised framework to integrate diverse multi-omics data (mixed data types and scales)
- ✓ Discovers latent factors capturing shared and modality-specific sources of variation
- ✓ Scales to large data sets and handles missing values
- ✓ Reveals biologically meaningful patterns linked to clinical subtypes and drug responses
- ✓ Open-source R package: MOFA

MOFA FRAMEWORK

MOFA uses a Bayesian probabilistic model (factor analysis) to explain the variation in each data set using a shared set of latent factors.



Model outputs

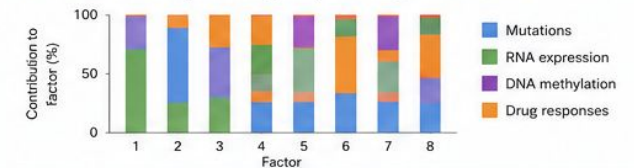


MOFA decomposes complex multi-omics data into biologically meaningful latent factors that summarize shared and specific patterns of variation.

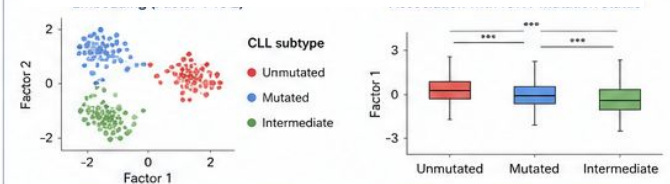
KEY RESULTS

1. Discovered latent factors capture major biology

The first few factors capture strong, interpretable signals across omics.

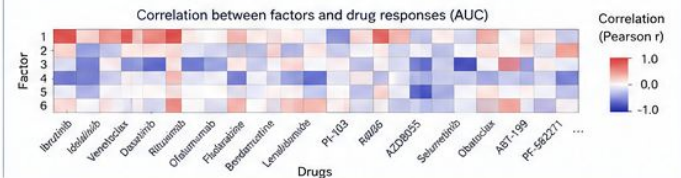


Mutated IGHV is generally linked to slower disease progression and better survival



3. Factors associated with drug responses

Specific factors correlate with ex vivo responses to targeted therapies.



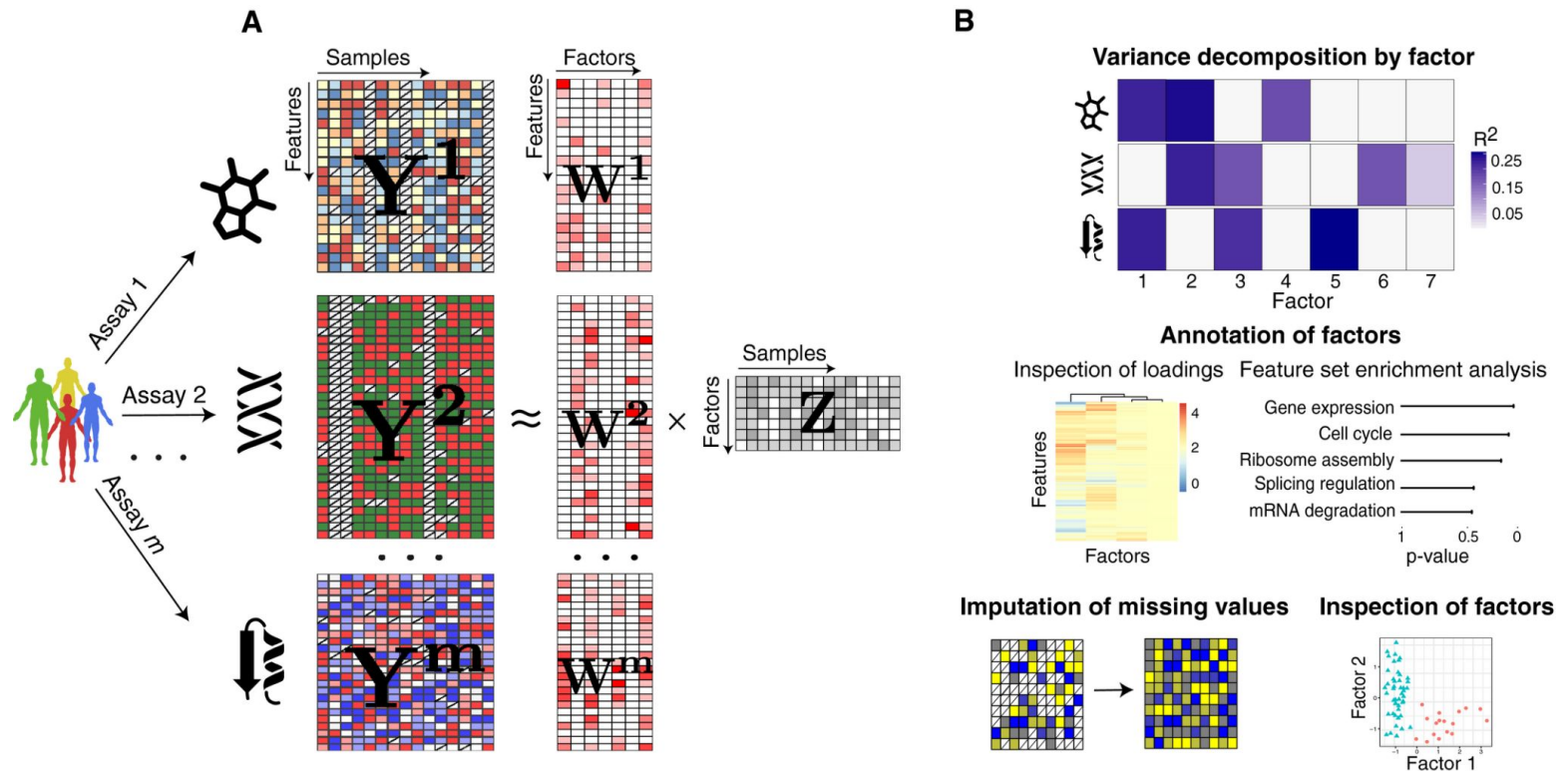
Factor 1 → IGHV mutation status (major CLL driver): explains 12% RNA variance, 38% DNA methylation variance

Factor 3 → trisomy of chromosome 12 (structural genomic change, active in mutations + RNA)

Factor 5 → oxidative stress response (previously underappreciated CLL driver)

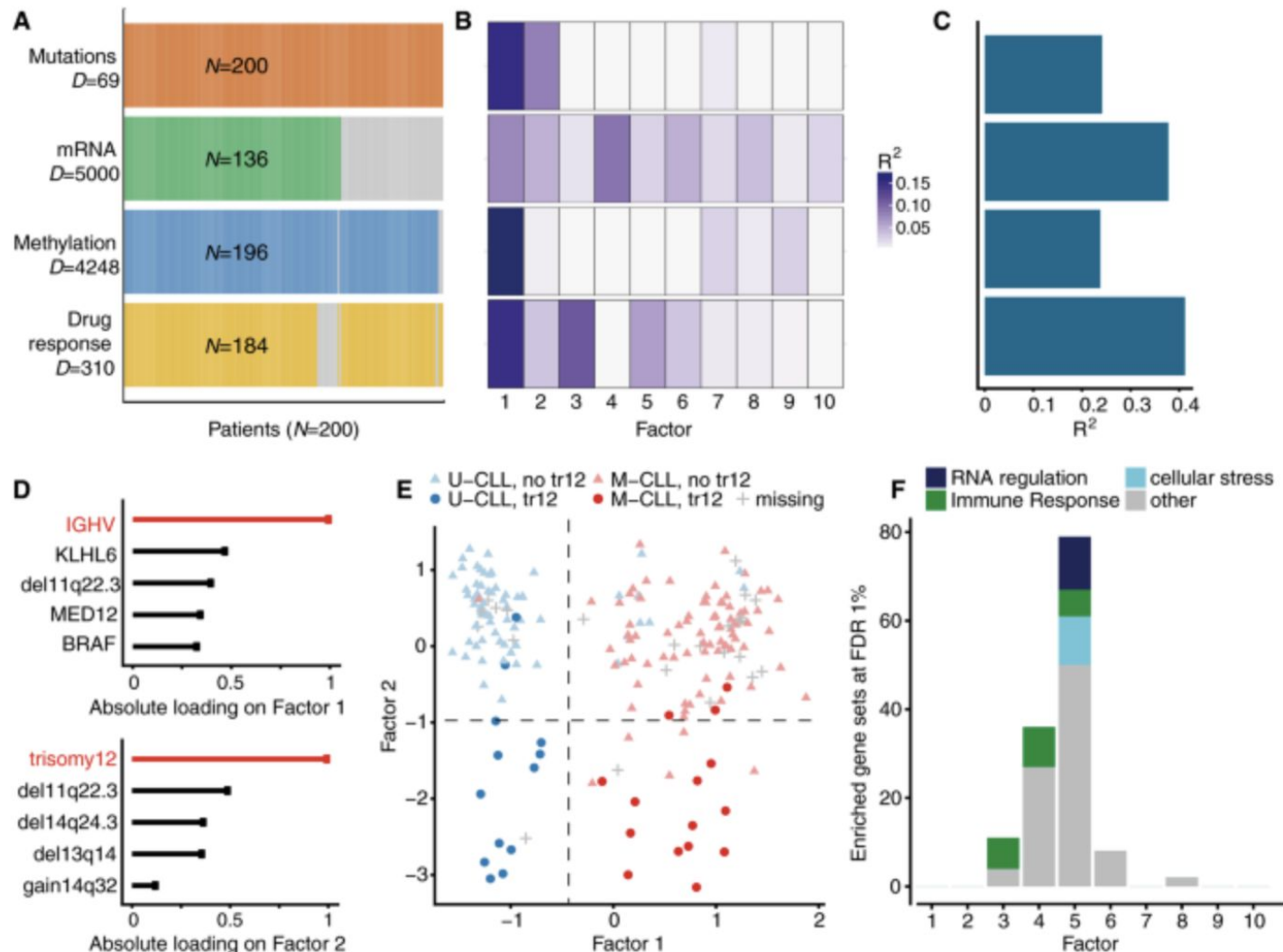
MOFA - original publication

Multi-Omics Factor Analysis - a framework for unsupervised integration of multi-omics data sets



MOFA - original publication

Multi-Omics Factor Analysis - a framework for unsupervised integration of multi-omics data sets



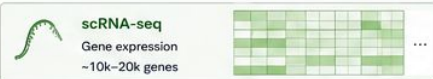
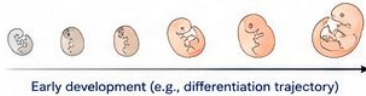
MOFA+: A Statistical Framework for Comprehensive Integration of Multi-Modal Single-Cell Data

Argelaguet et al., Genome Biology (2020) | ~700+ citations

MOFA+ extends MOFA to single-cell data, enabling scalable, flexible and interpretable integration of multiple data modalities across conditions and developmental stages.

STUDY DESIGN & DATA

Single-cell multi-modal data across developmental stages

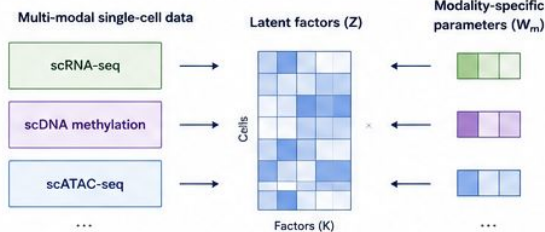


KEY CONTRIBUTIONS

- Introduces MOFA+, a scalable Bayesian framework for integrating multi-modal single-cell data.
- Uses variational inference for computational efficiency on large datasets.
- Supports flexible sparsity to learn interpretable factors.
- Models multiple sample groups and unbalanced data with missing modalities.
- Demonstrates biological insight by linking gene expression variation to DNA methylation changes during development.
- Open-source R package: MOFA2.

MOFA+ FRAMEWORK

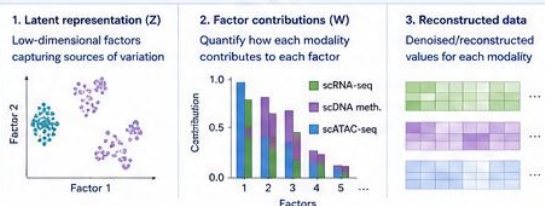
Unsupervised Bayesian deep generative model fitted with scalable variational inference



Key features

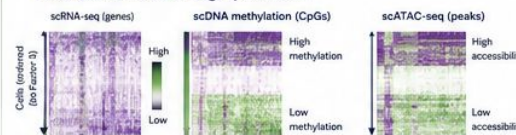


Outputs



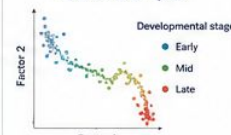
APPLICATION EXAMPLE: CONNECTING TRANSCRIPTOME TO EPIGENOME

Factor associated with lineage specification



Coordinated changes across modalities along the factor, reflecting regulatory programs driving cell fate decisions.

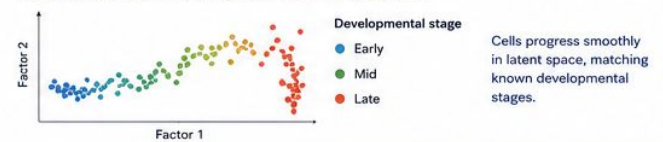
Factor 3 in latent space



KEY RESULTS

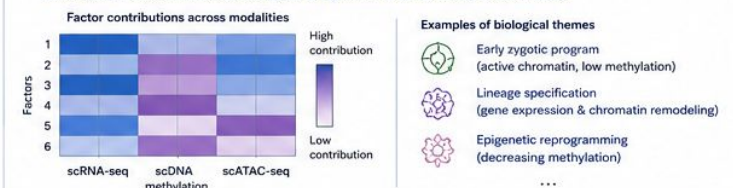
1. MOFA+ captures continuous biological variation across development

Latent factors order cells along developmental trajectories.



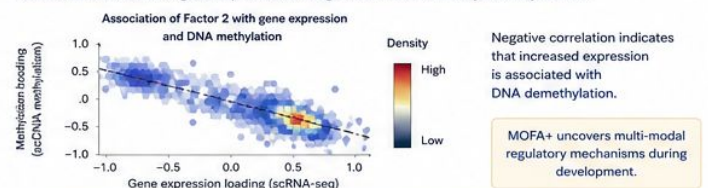
2. Factors reveal coordinated variation across modalities

Each factor reflects a distinct biological program shared by multiple modalities.



3. MOFA+ connects transcriptomic variation to DNA methylation changes

Factors associated with gene expression changes are linked to methylation dynamics.



IMPACT

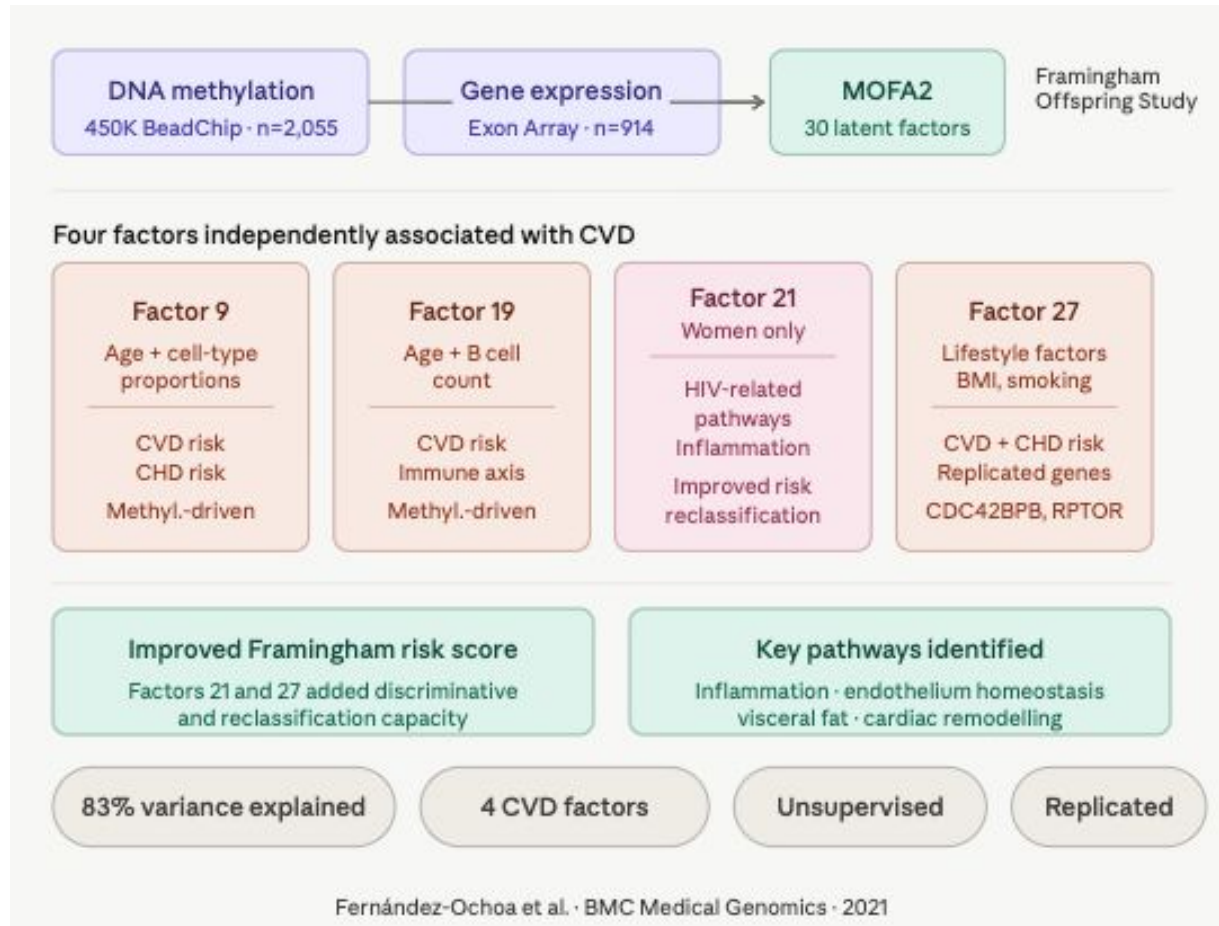
- Provides a powerful and flexible framework for comprehensive single-cell multi-omics integration.
- Enables discovery of shared and modality-specific regulatory programs.
- Facilitates mechanistic understanding of how epigenetic changes shape gene expression.
- Applicable to diverse biological questions and diseases.



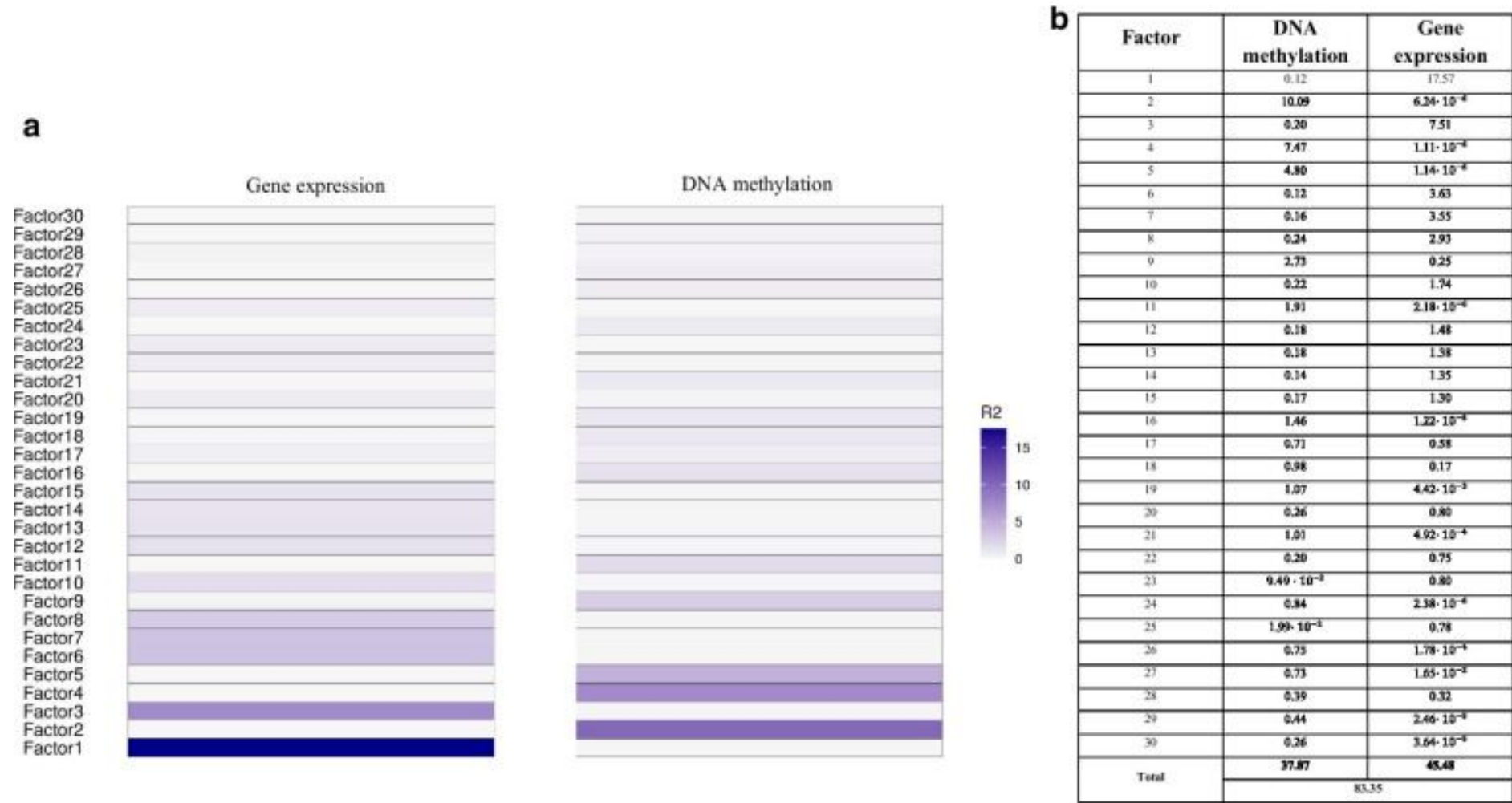
Take-home message: MOFA+ enables scalable, interpretable and biologically meaningful integration of multiple single-cell data types, revealing the coordinated interplay between the transcriptome, epigenome and chromatin during development.

GitHub: [bioFAM/MOFA2](https://github.com/bioFAM/MOFA2)
R package: MOFA2

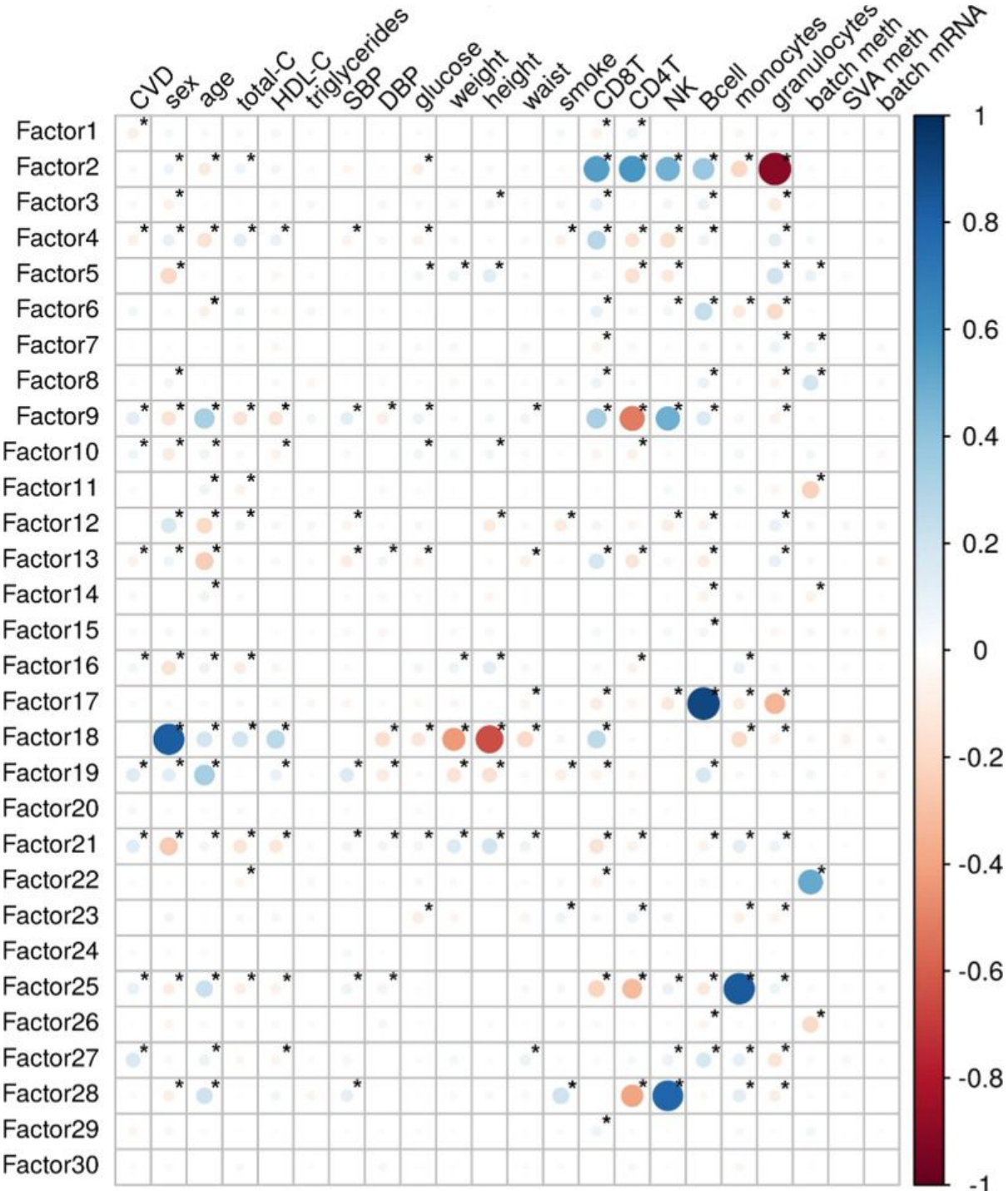
DNA methylation and gene expression integration in cardiovascular disease



DNA methylation and gene expression integration in cardiovascular disease



DNA methylation and
gene expression
integration in
cardiovascular disease



Comparing CCA and MOFA2

Aspect	CCA	MOFA2
Views	2 only	Any number
Regularization	None (overfits)	Automatic (Bayesian)
Feature selection	No (all weights non-zero)	Yes (sparse loadings)
Flexibility	Forces both views to contribute	Views contribute as needed
Uncertainty	Point estimates only	Full posterior distributions
Best use case	Strong signal in small datasets	Complex multi-omics studies

When to Use CCA vs MOFA2

Use CCA when:

- Two datasets only
- Strong prior belief both are relevant
- Goal: maximize correlation
- Small feature sets
- Exploratory phenotype association

Use MOFA2 when:

- Three or more omics layers
- Unknown which views are important
- Goal: robust pattern discovery
- Large feature sets (1000s)
- Need interpretable sparse loadings
- Require uncertainty quantification

Recommended workflow:

1. Start with MOFA2 as default
 - More principled, prevents overfitting
2. If signals are weak:
 - Check if CCA reveals stronger patterns
 - Validate with cross-validation
3. For publications:
 - MOFA2 preferred for multi-omics
 - Better justification for regularization
4. For prediction models:
 - Compare both, select by performance

Key Takeaways

Multi-omics integration is essential for systems biology:

- Reveals coordinated changes across molecular layers

- Identifies drivers of biological phenotypes

CCA: Classic approach for two-view correlation maximization

- Simple, interpretable, forces integration

- Good for exploratory analysis with small datasets

MOFA2: Modern Bayesian multi-view framework

- Handles any number of omics layers

- Automatic regularization and feature selection

- Best practice for robust multi-omics studies

Choice depends on context:

- Dataset size, number of views, analysis goals

Resources & Further Reading

MOFA2 Official Resources:

Website: <https://biofam.github.io/MOFA2/>

Publication: Argelaguet et al., Genome Biology (2020)

R package: `BiocManager::install('MOFA2')`

Python package: `pip install mofapy2`

CCA Background:

Hotelling (1936): Relations between two sets of variates

`sklearn.cross_decomposition.CCA` (Python)

Tutorials:

MOFA2 vignettes: <https://biofam.github.io/MOFA2/tutorials.html>

https://raw.githubusercontent.com/bioFAM/MOFA2_tutorials/master/R_tutorials/CLL.html

Multi-omics integration reviews: Subramanian et al. (2020)